S.No: 1	Exp. Name: ***Student Management System class***	Date:2026-10-17

Aim:

Create a Student Management System class to manage the student records. This class should have the following functionalities:

- Add a new student.
- Display all student information.
- Update student details (name, age, or grade) based on student ID.
- Delete a student based on student ID.

Note: Refer to the visible test cases for input and output formats

Source Code:

q32557/StudentManagementSystem.java

```
package q32557;
import java.util.*;

class Student {
    private int id;
    private String name;
    private int age;
    private double grade;

    // Constructor
    public Student(int id, String name, int age, double grade) {
        this.id = id;
        this.name = name;
        this.age = age;
        this.grade = grade;
    }

    // Getters and Setters
    public int getId() {
        return id;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public int getAge() {
        return age;
    }

    public void setAge(int age) {
        this.age = age;
    }
```


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```
        return grade;
    }

    public void setGrade(double grade) {
        this.grade = grade;
    }

    // Method to display student information
    public void display() {

        System.out.println("Student{id=" + id + ", name='" + name + "', age=" + age +
", grade=" + grade + "}");// Write your code here

    }
}

public class StudentManagementSystem {
    private Student[] students;
    private int size; // current number of students in the system

    // Constructor to initialize the system
    public StudentManagementSystem(int capacity) {
        this.students = new Student[capacity];
        this.size = 0;
    }

    // Method to add a new student
    public void addStudent(int id, String name, int age, double grade) {

       if (size < students.length) {
            students[size++] = new Student(id, name, age, grade);
           }//Write your code here

    }

    // Method to display all students
    public void displayAllStudents() {

        if (size == 0) {
            System.out.println("No students to display");
            return;
        }
        for (int i = 0; i < size; i++) {
            students[i].display();
                }// Write your code here

    }

    // Method to find a student index by ID
    private int findStudentIndexById(int id) {

        for (int i = 0; i < size; i++) {
            if (students[i].getId() == id) {
                return i;
            }
        }
```

```
        return -1;// Write your code here

    }

    // Method to update student details by ID
    public void updateStudentDetails(int id, String newName, int newAge, double newGr
ade) {

        int index = findStudentIndexById(id);
        if (index != -1) {
            students[index].setName(newName);
            students[index].setAge(newAge);
            students[index].setGrade(newGrade);
            System.out.println("Updated successfully");
        } else {
            System.out.println("ID " + id + " not found");
        }// Write your code here

    }

    // Method to delete a student by ID
    public void deleteStudent(int id) {

        int index = findStudentIndexById(id);
        if (index != -1) {
            for (int j = index; j < size - 1; j++) {
                students[j] = students[j + 1];
            }
            students[--size] = null;
            System.out.println(id + " deleted successfully");
        } else {
            System.out.println("ID " + id + " not found");
                }// Write your code here

    }

    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        //System.out.println("Student Management System Menu:");
        System.out.println("1. Add a new student");
        System.out.println("2. Display all students");
        System.out.println("3. Update student details");
        System.out.println("4. Delete a student");
        System.out.println("5. Exit");

        StudentManagementSystem sms = new StudentManagementSystem(100);// Write code
to set a capacity for 100 students

        while (true) {
            System.out.print("Choice: ");
            int choice = scanner.nextInt();
            scanner.nextLine(); // consume newline

            switch (choice) {
                case 1:
                    System.out.print("ID: ");
```

```
                    int id = scanner.nextInt();
                    scanner.nextLine(); // consume newline
                    System.out.print("Name: ");
                    String name = scanner.nextLine();
                    System.out.print("Age: ");
                    int age = scanner.nextInt();
                    System.out.print("Grade: ");
                    double grade = scanner.nextDouble();
                    sms.addStudent(id, name, age, grade);
                    break;
                case 2:
                    sms.displayAllStudents();
                    break;
                case 3:
                    System.out.print("ID: ");
                    int updateId = scanner.nextInt();
                    scanner.nextLine(); // consume newline
                    System.out.print("New name: ");
                    String newName = scanner.nextLine();
                    System.out.print("New age: ");
                    int newAge = scanner.nextInt();
                    System.out.print("New grade: ");
                    double newGrade = scanner.nextDouble();
                    sms.updateStudentDetails(updateId, newName, newAge, newGrade);
                    break;
                case 4:
                    System.out.print("ID: ");
                    int deleteId = scanner.nextInt();
                    sms.deleteStudent(deleteId);
                    break;
                case 5:
                    scanner.close();
                    System.exit(0);
                default:
                    System.out.println("Invalid choice");
            }

            // System.out.println("Student Management System Menu:");
            System.out.println("1. Add a new student");
            System.out.println("2. Display all students");
            System.out.println("3. Update student details");
            System.out.println("4. Delete a student");
            System.out.println("5. Exit");
        }
    }
}
```

Execution Results - All test cases have succeeded!

Test Case - 1
User Output
1. Add a new student
2. Display all students
3. Update student details

```
4. Delete a student
5. Exit
Choice:
1
ID:
101
Name:
Camio
Age:
25
Grade:
83.2
1. Add a new student
2. Display all students
3. Update student details
4. Delete a student
5. Exit
Choice:
3
ID:
101
New name:
Alice
New age:
26
New grade:
90
Updated successfully
1. Add a new student
2. Display all students
3. Update student details
4. Delete a student
5. Exit
Choice:
2
Student{id=101, name='Alice', age=26, grade=90.0}
1. Add a new student
2. Display all students
3. Update student details
4. Delete a student
5. Exit
Choice:
1
ID:
102
Name:
Ram
Age:
30
Grade:
```

95
1. Add a new student
2. Display all students
3. Update student details
4. Delete a student
5. Exit
Choice:
2
Student{id=101, name='Alice', age=26, grade=90.0}
Student{id=102, name='Ram', age=30, grade=95.0}
1. Add a new student
2. Display all students
3. Update student details
4. Delete a student
5. Exit
Choice:
4
ID:
102
102 deleted successfully
1. Add a new student
2. Display all students
3. Update student details
4. Delete a student
5. Exit
Choice:
2
Student{id=101, name='Alice', age=26, grade=90.0}
1. Add a new student
2. Display all students
3. Update student details
4. Delete a student
5. Exit
Choice:
5

Test Case - 2
User Output
1. Add a new student
2. Display all students
3. Update student details
4. Delete a student
5. Exit
Choice:
3
ID:
103
New name:
Krish
New age:

5
New grade:
89
ID 103 not found
1. Add a new student
2. Display all students
3. Update student details
4. Delete a student
5. Exit
Choice:
2
No students to display
1. Add a new student
2. Display all students
3. Update student details
4. Delete a student
5. Exit
Choice:
4
ID:
103
ID 103 not found
1. Add a new student
2. Display all students
3. Update student details
4. Delete a student
5. Exit
Choice:
5